

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS student (NC State, May 2026) with production experience building AI/ML systems across fintech, healthcare, and enterprise environments. Built Glance AI -- a compliance-ready RAG system with per-tenant data isolation for fintech and healthcare, and PerCent, a GenAI finance app with RAG-based chatbot ranked top 10 of 100+ teams at HackNC24. At Mphasis, built OCR/NLP/ML automation workflows that improved health claim accuracy by 25%. Published IEEE research on a T5 transformer pipeline (Mapify) processing 1,500+ documents. Strong in Python, PyTorch, TensorFlow, distributed data pipelines (Kafka, Spark), and end-to-end ML from research through production.

CORE COMPETENCIES

Python (ML & Scripting)

PyTorch & TensorFlow

End-to-End ML Pipelines

RAG & LLM Integration

Distributed Data Pipelines (Kafka / Spark)

Fintech & Compliance AI

Real-Time APIs & Decisioning

Data Annotation & Model Evaluation

Production AI Deployment

EXPERIENCE

Mphasis (HP Company)

Jun 2024 - Aug 2024

Software Developer Intern

- Built **5+ AI-driven automation workflows** in Python combining OCR, NLP, and ML classification models to automate health insurance claim processing -- improved claim accuracy by **25%**
- Designed and deployed ML validation pipelines on AWS & GCP; integrated Grafana dashboards for model performance monitoring and drift detection
- Automated model red-teaming pipelines to stress-test edge cases before production release

Isomer AI

May 2025 - Present

Product Engineer Intern - Remote

- Integrated **multi-model LLM pipelines** across Claude, GPT-4o, and Gemini using MCP (Model Context Protocol) for structured tool-calling, function execution, and AI reasoning workflows in a production claims automation platform
- Built real-time event pipelines, document processing systems (PDF/email/S3), and RBAC -- **394 commits** across 7 production deployments on TypeScript/Node.js/PostgreSQL/Redis stack
- Integrated Temporal.io for durable workflow orchestration; monitored deployed models with Sentry observability across all production environments

NC State University

Apr 2025 - Present

Graduate Research Assistant

- Applied **hyperparameter tuning and time-series ML** for genomic classification models, improving accuracy by 12%; built a cloud-based data platform on GCP serving 100+ researchers across 500GB datasets

Capgemini

Jun 2023 - Aug 2023

AI/ML Intern

- Built **Kafka streaming + Spark batch pipelines** for real-time support ticket classification and analysis; reduced pipeline latency by 20% and processing time by 15%

PROJECTS

Glance AI - Fintech & Healthcare RAG System Independent · 2025

Built a **compliance-ready RAG system** for fintech and healthcare environments with **per-tenant data isolation at the vector retrieval layer** -- 0 cross-tenant context leakage by design. Deterministic retrieval architecture eliminates hallucinations for regulated data workflows.

Python · LLMs · Vector Databases · RAG · Compliance-ready architecture

PerCent - GenAI Finance App HackNC24 Top 10 of 100+ teams

Full-stack **fintech app** with goal tracking, expense logging, and a **RAG-powered AI chatbot** (Qwen-14B-Chat) reducing manual financial research by 30%. Built complete MVP in 48 hours using Python, Hugging Face, and Selenium for data ingestion -- ranked top 10 at HackNC24.

Python · Hugging Face · Qwen-14B-Chat · RAG · Selenium · React

AlgoPulse - Distributed AI Orchestration Platform Independent · 2025

Designed an event-driven AI platform, identifying a polling bottleneck and re-architecting to **Temporal.io durable workflows** + Kafka state sync + Supabase Edge Function streaming. Scaled to **2,000 async AI inference requests/second** with zero state loss under fault conditions.

Python · TypeScript · Node.js · Temporal.io · Kafka · Supabase Edge Functions

Mapify - ML & NLP Pipeline IEEE Published

T5-transformer model pipeline that converted 1,500+ PDFs and images into structured mind maps and flowcharts, reducing academic review time by 60%. Presented to 500+ attendees at IEEE conference.

Python · T5 Transformer · PyTorch · NLTK · NetworkX

EDUCATION

North Carolina State University - MS Computer Science Aug 2024 - May 2026

Courses: Artificial Intelligence, Data Science, Machine Learning, Software Engineering, Algorithms

Mumbai University - BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 · Courses: DS & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

AI/ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain, RAG, OpenAI API, Anthropic Claude, MCP, Pandas, NumPy

Languages: Python, TypeScript, JavaScript, Java, SQL, C, R

Data Pipelines: Kafka, Spark, Hadoop, Temporal.io, Redis, Supabase Edge Functions

Cloud & Infra: AWS, GCP, Docker, CI/CD, Grafana, Sentry

Web: Node.js, React, Next.js, Express, PostgreSQL, Git